Antika Roy

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SUMMARY

Graduate research assistant with 3.5 years of experience in deep learning, computer vision, LLM and generative AI with publications in MICCAI, ECTC, SPIE, ACM, IEEE conferences, one tech-license, and two patents.

EDUCATION

Ph.D. in Electrical and Computer Engineering Expected Graduation: *December 2027*
University of Florida, Gainesville, FL, USA *CGPA: 3.70/4.00*
Master of Science in Electrical and Computer Engineering *May 2026*
University of Florida, Gainesville, FL, USA *CGPA: 3.70/4.00*

SKILLS

Programming & Tools: Python, Java, C/C++, Git, GitHub, MATLAB
AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV
Machine Learning Expertise: Agentic AI, Vison Language Models, Large Language Models (Fine-Tuning with LoRA/QLoRA, Evaluation & Benchmarking), Explainable AI, Physics-Informed AI, Diffusion Model, Generative AI
Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn, SQL (SQLite, PostgreSQL, SQLAlchemy), NoSQL (MongoDB, ChromaDB)
Cloud & Compute Acceleration: AWS

WORK EXPERIENCE

U.S. Permanent Resident (Fully authorized to work in the U.S. without sponsorship)

Graduate Research Assistant *May 2023 – Present*
SCAN Lab, Electrical and Computer Engineering, University of Florida *Gainesville, FL, USA*
Director: Dr. Navid Asadi

- Developing **deep learning-based** and **multimodal systems** for **PCB/advanced packaging inspection** and **reverse engineering** using **PyTorch, TensorFlow**, **OpenCV**. Authored **32 peer-reviewed papers** in **MICCAI/ACM/IEEE/SPIE** venues through **cross-disciplinary research collaborations**.

Java Developer *September 2019 – December 2020*
Wipro Limited *Dhaka, Bangladesh*

- Developed **backend APIs** for **GPAY App** using Java EE, Rest, SOAP, jQuery, JavaScript, MySQL.

Junior Engineer *July 2018 – August 2019*
Swosti Limited *Dhaka, Bangladesh*

- Implemented microcredit business logic for **an automated microcredit and micro-savings management application** using Java SE, MySQL procedure & functions, AWS EC2, AWS RDS, SVN.

PROJECT HIGHLIGHTS

Agentic AI,VLM and LLM Assistant for Automated Inspection *June 2025 - Present*

- Developed a **cross-context RAG framework** by combining hybrid retrieval, query decomposition, and multi-hop evidence gathering, resulting in **0.619 document-level evidence recall and improved QA accuracy and interpretability** for photonics packaging inspection and training.
- Developed AURA-PIE, an AI-enabled retrieval-augmented assistant for electronics inspection, integrating domain-specific QA data, LoRA fine-tuning, and RAG to improve F1 score by 75.7% over fine-tuned baseline.

Physics Informed Neural Network for Image Reconstruction
May 2025

- Formulated a physics-informed neural network framework to enhance noisy SAM images for IC packaging inspection, achieving strong denoising performance with 69.66 dB PSNR, 0.4815 SSIM, and physics consistency with PCRS of 1.00e-08.

Wavelet-driven Feature Representation of Medical Image Segmentation
July 2025

- Developed **WaveFormer**, a wavelet-driven 3D Transformer for efficient global–local feature modeling and achieved state-of-the-art 3D medical image segmentation with **~91.4%** mean Dice on **BraTS2023** and **~93.8%** on **FLARE2021 dataset**.

Deep Learning for PCB Component Analysis and Inspection *May 2023 – August 2025*

- Increased accuracy of SAM for PCB component segmentation by **PEFT** on limited annotated X-ray dataset with improved segmentation with **patch-based data augmentation** techniques, **boosting IoU by ~34%**.

AWARDS AND CERTIFICATIONS

Awarded Intel Best Student Paper *in the IEEE 75th Electronic Components and Technology Conference*

NVIDIA-Certified Associate: Generative AI LLMs *December 2025*

PATENTS & TECH LICENSES

Image Reconstruction with Multimodal Fusion and Physics-Informed Neural Network *Filed: July 21, 2025*

Explainable AI Metrics for IC Packaging Inspection *Filed: July 8, 2025*

SegPCBX: X-ray Dataset for Automated PCB Component Analysis and Quality Assurance (Tech License)